DPP No: 33

Maximum Time 50 Min

competishun Power of real gurus

C³

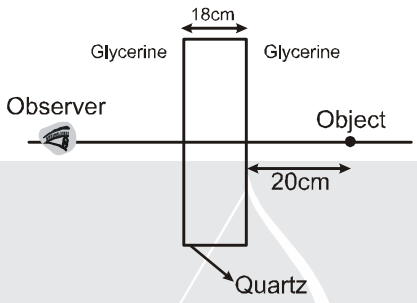
(COMPETISHUN CRASH COURSE)

PHYSICS

TARGET JEE-MAINS

SYLLABUS : GEOMETRICAL OPTICS

1. Given that, velocity of light in quartz = 1.5×10^8 m/s and velocity of light in glycerine = $(9/4) \times 10^8$ m/s. Now a slab made of quartz is placed in glycerine as shown. The shift of the object produced by slab is

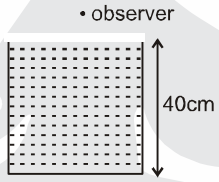


(A) 6 cm (B) 3.55 cm (C) 9 cm (D) 2 cm

2. The critical angle of light going from medium A to medium B is θ. The speed of light in medium A is v. The speed of light in medium B is:

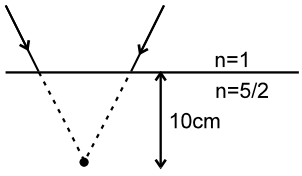
(A) $\frac{v}{\sin\theta}$ (B) $v \sin\theta$ (C) $v \cot\theta$ (D) $v \tan\theta$

3. In the given figure an observer in air (n = 1) sees the bottom of a beaker filled with water (n = 4/3) upto a height of 40 cm. What will be the depth felt by this observer.



(A) 20 cm (B) 30 cm (C) 50 cm (D) 90 cm

4. In the given figure rays incident on an interface would converge 10 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet some where else. Find the distance of meeting point of refracted rays below the interface, assuming the rays to be making small angles with the normal to the interface.



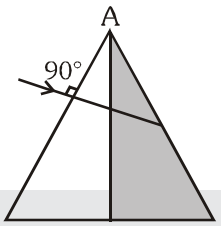
(A) 10 cm (B) 15 cm (C) 25 cm (D) 30 cm

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5. Light goes from glass ($\mu = \frac{3}{2}$) to air. Find the angle of incidence for which the angle of deviation is 90º.

(A) 15° (B) 25° (C) 30° (D) 45°

6. A ray of light is incident normally on the first refracting face of the prism of refracting angle A. The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will :–



(A) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/2\right)$

(B) Not emerge out of the prism

(C) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec A/4\right)$

(D) None of these

7. A ray of monochromatic light is incident on one refracting face of a prism of angle 75^0. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is

(A) 30^0 (B) 45^0 (C) 60^0 (D) 0^0

8. A prism having refractive index $\sqrt{2}$ and refracting angle 30º, has one of the refracting surfaces polished. A beam of light incident on the other refracting surface will retrace its path if the angle of incidence is:

(A) 0º (B) 30º (C) 45º (D) 60º

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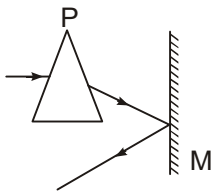
9. A ray of light is incident at angle *i* on a surface of a prism of small angle A and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, the angle of incidence *i* is nearly equal to :

(A) A/μ (B) $A/(2\mu)$ (C) μA (D) $\mu A/2$

10. The maximum refractive index of a material, of a prism of apex angle 90º, for which light may be transmitted is:

(A) $\sqrt{3}$ (B) 1.5 (C) $\sqrt{2}$ (D) None of these

11. A prism having an apex angle of 4^0 and refractive index of 1.50 is located in front of a vertical plane mirror as shown in the figure. A horizontal ray of light is incident on the prism. The total angle through which the ray is deviated is:



(A) 4^0 clockwise (B) 178^0 clockwise (C) 2^0 clockwise (D) 8^0 clockwise

12. The cross section of a glass prism has the form of an equilateral triangle. A ray is incident onto one of the faces perpendicular to it. Find the angle θ between the incident ray and the ray that leaves the prism. The refractive index of glass is $\mu = 1.5$.

(A) 10° (B) 30° (C) 45° (D) 60°

13. A concave spherical surface of radius of curvature 10 cm, separates two medium X and Y of R.I. 4/3 and 3/2 respectively. If the object is placed along principal axis in medium X , then :



(A) image is always real

(B) image is real if the object distance is greater than 90 cm

(C) image is always virtual

(D) image is virtual if the object distance is less than 90 cm

14. There is a small black dot at the centre C of a solid glass sphere of refractive index μ. When seen from outside, the dot will appear to be located:

(A) away from C for all values of μ

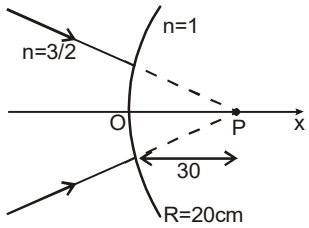
(B) at C for all values of μ

(C) at C for $\mu = 1.5$, but away from C for $\mu \neq 1.5$

(D) at C only for $\sqrt{2} \leq \mu \leq 1.5$.

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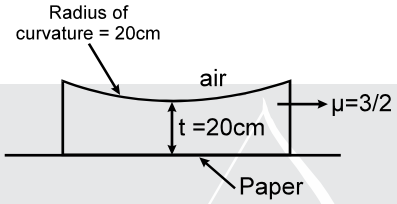
15. The image for the converging beam after refraction through the curved surface (in the given figure) is formed at:



(A) x = 40 cm (B) x = $\frac{40}{3}$ cm

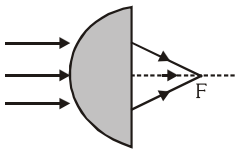
(C) x = $-\frac{40}{3}$ cm (D) x = $\frac{180}{7}$ cm

16. In the given figure a plano-concave lens is placed on a paper on which a flower is drawn. How far above its actual position does the flower appear to be ?



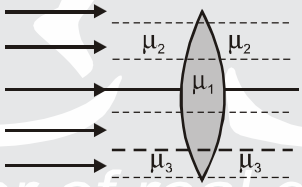
(A) 10 cm (B) 15 cm (C) 50 cm (D) none of these

17. A paraxial beam is incident on a glass (n = 1.5) hemisphere of radius R=6 cm in air as shown. The distance of point of convergence F from the plane surface of hemisphere is :–



(A) 12 cm (B) 5.4 cm (C) 18 cm (D) 8 cm

18. A double convex lens, made of a material of refractive index μ_1, is placed inside two liquids of refractive indices μ_2 and μ_3, as shown. $\mu_2 > \mu_1 > \mu_3$. A wide, parallel beam of light is incident on the lens from the left. The lens will give rise to :–



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(A) a single convergent beam (B) two different convergent beams

(C) two different divergent beams (D) a convergent and a divergent beam

19. A convexo - concave diverging lens is made of glass of refractive index 1.5 and focal length 24 cm. Radius of curvature for one surface is double that of the other. Then radii of curvature for the two surfaces are (in cm):

(A) 6, 12 (B) 12, 24 (C) 3, 6 (D) 18, 36

20. Two symmetric double convex lenses A and B have same focal length, but the radii of curvature differ so that $R_A = 0.9\ R_B$. If $n_A = 1.63$, find n_B.

(A) 1.7 (B) 1.6 (C) 1.5 (D) 4/3

21. When a lens of power P (in air) made of material of refractive index μ is immersed in liquid of refractive index μ_0. Then the power of lens is:

(A) $\frac{\mu - 1}{\mu - \mu_0}$ P (B) $\frac{\mu - \mu_0}{\mu - 1}$ P (C) $\frac{\mu - \mu_0}{\mu - 1} \cdot \frac{P}{\mu_0}$ (D) none of these

22. A lens behaves as a converging lens in air and a diverging lens in water. The refractive index of the material is (refractive index of water = 1.33)

(A) equal to unity (B) equal to 1.33

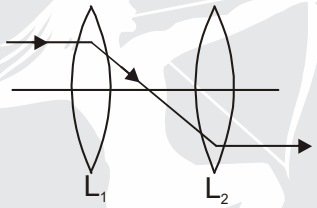
(C) between unity and 1.33 (D) greater than 1.33

23. A thin symmetrical double convex lens of power P is cut into three parts, as shown in the figure. Power of A is:



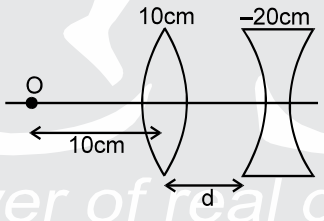
(A) 2 P (B) $\frac{P}{2}$ (C) $\frac{P}{3}$ (D) P

24. In the figure given below, there are two convex lens L_1 and L_2 having focal length of f_1 and f_2 respectively. The distance between L_1 and L_2 will be



(A) f_1 (B) f_2 (C) $f_1 + f_2$ (D) $f_1 - f_2$

25. What should be the value of distance d so that final image is formed on the object itself. (focal lengths of the lenses are as given in the figure).



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(A) 10 cm (B) 20 cm (C) 5 cm (D) none of these

ANSWER KEY

1.	(A)	**2.**	(A)	**3.**	(B)	**4.**	(C)	**5.**	(D)
6.	(A)	**7.**	(B)	**8.**	(C)	**9.**	(C)	**10.**	(C)
11.	(B)	**12.**	(D)	**13.**	(C)	**14.**	(B)	**15.**	(A)
16.	(A)	**17.**	(D)	**18.**	(D)	**19.**	(A)	**20.**	(A)
21.	(C)	**22.**	(C)	**23.**	(D)	**24.**	(C)	**25.**	(A)